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Unit 3 Designing Solutions- Utilizing "While" Loop

PDF Documents

PDF Document 1: 1731843195588-Chapter 4_ unit3 Designing Solutions.pdf

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UNIT 3 DESIGNING SOLUTIONS

● Programming Concepts

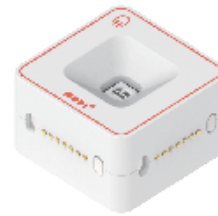
WHILE LOOP

A "while" loop continues to execute its block of code **as long as the condition is true**. It checks the condition before running the loop's code. If the condition is true, the loop runs. If it's false, the loop stops. For example, "while the sun is up, keep the LED off."

✓ WHILE



condition
While the sun is up,



task
Keep the LED off

Another Example:

"Think of using a 'while' loop like brushing your teeth while a timer is running. You start brushing, and 'while' the timer hasn't beeped (meaning time is not up yet), you keep brushing. As soon as the timer beeps, signaling the time is up, you stop brushing."



condition
While the timer is running,



task
Keep brushing your teeth.

**WAIT**

The wait block is like a pause button for your code. It stops the code from moving on to the next step for a while. So, if a motor is spinning or an LED is shining, they'll keep doing their thing during the pause. The wait block just makes sure the code waits before jumping to the next action.

**DID YOU KNOW?**

When using the wait block, it's important to remember: while the wait block is running, no other commands are executed. No buttons can be pressed, no dials can be turned, and all the code will stop



Let's read the flowchart and create the code.

FLOWCHART

